EAST WEST UNIVERSITY

MINI PROJECT REPORT

Design a full-fledged network for an organization with multiple subnets.

CSE405: Computer Networks
Section 01
Summer 2025

Prepared for

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Introduction

Apex University, similar to East West University, has many computers and a large network system. Every classroom, lab, office, library, and other academic or administrative section has wired internet, while the whole campus also has wireless internet. The university uses different connected systems to manage important tasks such as admissions, advising, results, calendar, library services, accounts, and more.

Our job is to design a full network model that shows how all these systems and subnetworks are linked together, matching the structure and services of the university.

Purpose of network

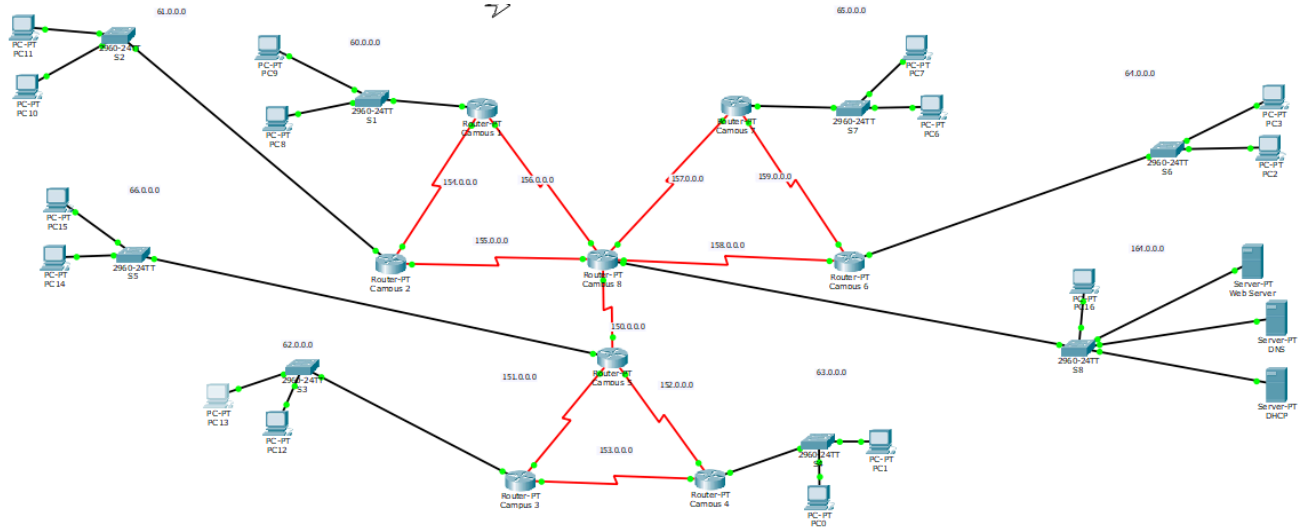
- Provide safe and reliable internet access for all eight campuses.
- Use dynamic routing and DHCP to keep the network connected and manageable.
- Support web hosting, DNS services, and smooth communication between devices.
- Design the network so it can grow and expand in the future.

Design Specification

The network design uses eight routers, one for each campus. Each campus has its own networks, based on Class A addresses. A switch connects each router to the campus network. From the switch, different sections like classrooms, labs, staff PCs, the library, and other departments are connected. Each campus also has a wireless network, with access points linked to the switch.

Campus 8 has a server room that uses a Class B network. This server room includes the DHCP, DNS, and Web servers. A single DHCP server provides IP addresses for all campuses. The web server hosts the university's main website, which handles services like admissions, advising, results, calendar, library management, and accounts. The links between the routers use Class B addresses.

Physical Diagram



Design Issues :

Number of hosts: 15 PCs

Number of networks:

LAN Segments (Local Area Networks): These are the networks to which the PCs and servers are directly connected.

- In the topology, there are **7** LAN segments with PCs attached (behind Routers 1, 2, 3, 4, 5, 6, and 7).
- There is **1** additional LAN segment with the servers and PC 16.
- This gives you a total of 7+1=8 LAN segments.

WAN Segments (Point-to-Point Links): These are the links connecting your routers. Each link between two routers is its own network.

- In the image, there are **10** such links. You can count them by following the red lines between the routers.
- These links are typically configured as small subnets, often with only two usable addresses (one for each router's interface).

Total 8 + 10 = 18 number of networks

Limitations and problems faced :

If the DNS server is not added in the DHCP pool, the devices cannot access the web server through dns. So I checked using “show running-config” command if dns was added in the pool. Otherwise I had to remove the pool using “no ip dhcp pool <poolname>” command and properly add the dhcp pool. Also to ensure everything goes smoothly I had to use write erase and reload command multiple times in EXEC mode of router (router#) and eventually when everything went right there were still some unexpected issues in the network like website loading slowly or not loading and various bugs but after a quick restart and cleaning my windows IP I was able to get rid of this software bug and now in every PC the website is running smoothly and I also hope that this pkt file runs smoothly in other devices as well . I also had to start over multiple times just because I pressed the wrong command or if I was careless for a moment . Even though it was a mini project it was very attention to detail intensive which was very tough to execute properly and without repetitive effort.

Codes:

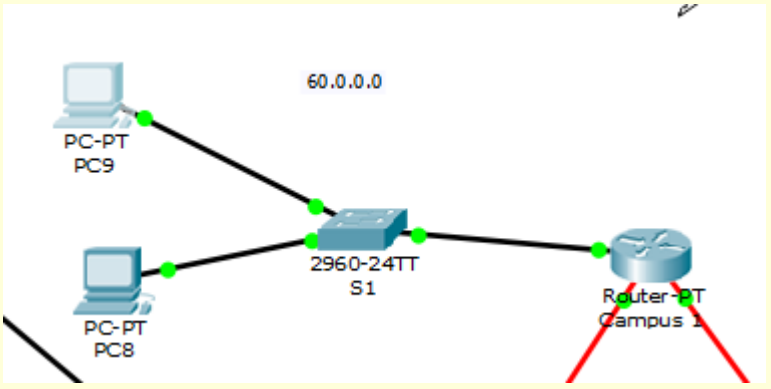
Routing Codes

Campus 1

```
interface fa0/0
ip address 60.0.0.254 255.0.0.0
no shut
do wr
exit

interface se2/0
ip address 156.0.0.1 255.255.0.0
clock rate 64000
no shut
do wr
exit

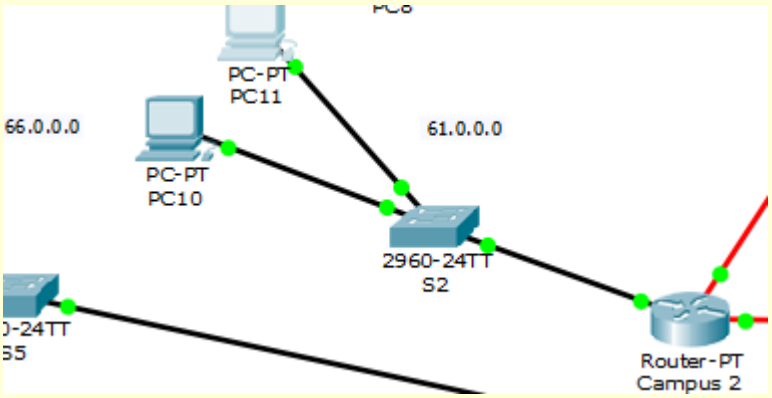
interface se3/0
ip address 154.0.0.1 255.255.0.0
clock rate 64000
no shut
do wr
Exit
```



Campus 2

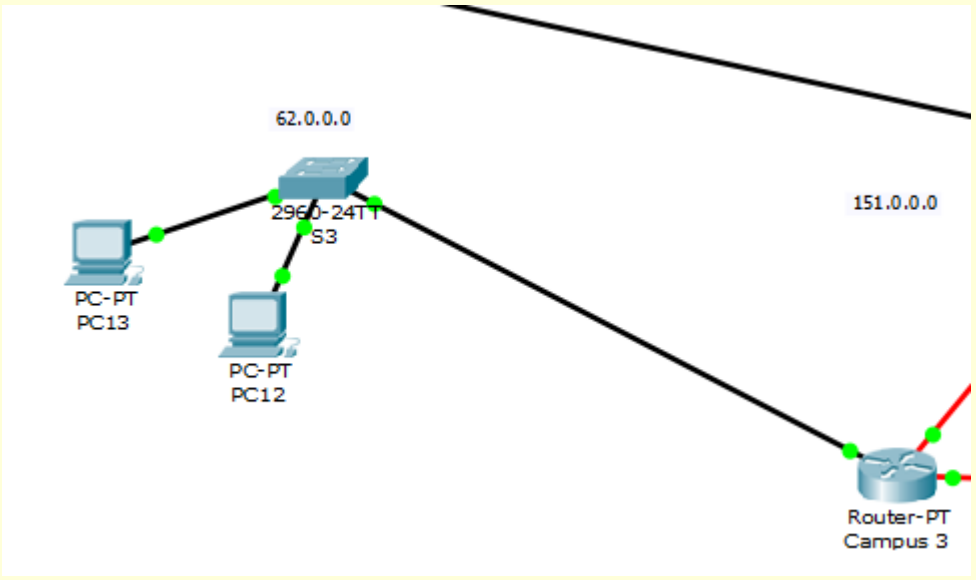
```
interface fa0/0
ip address 61.0.0.254 255.0.0.0
no shut
do wr
exit
interface se2/0
ip address 155.0.0.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 154.0.0.2 255.255.0.0
```

```
no shut
do wr
exit
```



Campus 3

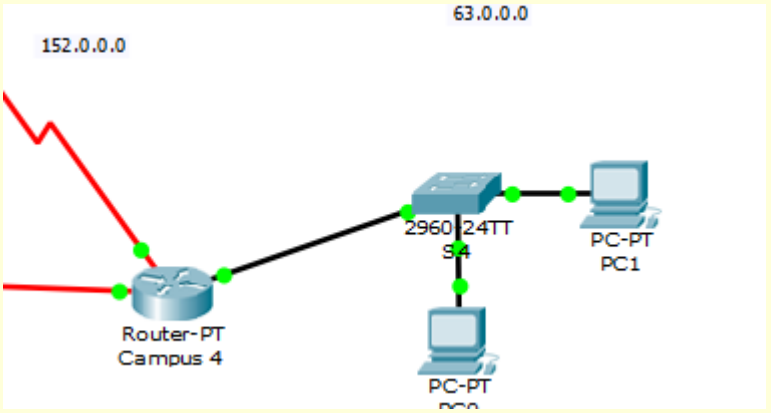
```
interface fa0/0
ip address 62.0.0.254 255.0.0.0
no shut
do wr
exit
interface se2/0
ip address 151.0.0.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 153.0.0.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
```



Campus 4

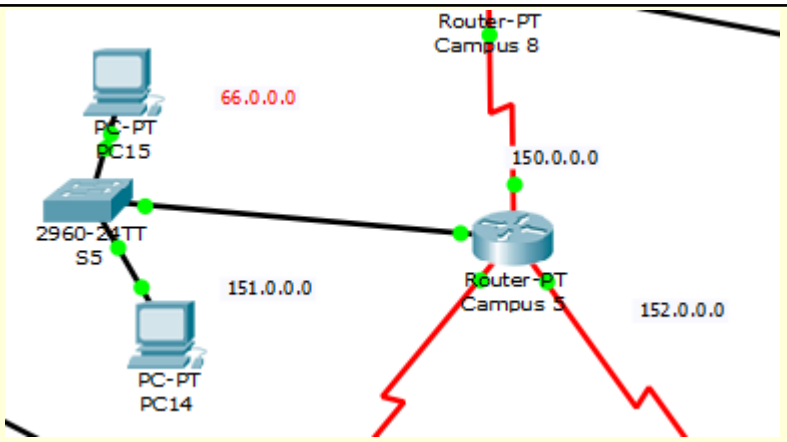
```
interface fa0/0
```

```
ip address 63.0.0.254 255.0.0.0
no shut
do wr
exit
interface se2/0
ip address 152.0.0.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 153.0.0.2 255.255.0.0
no shut
do wr
Exit
```



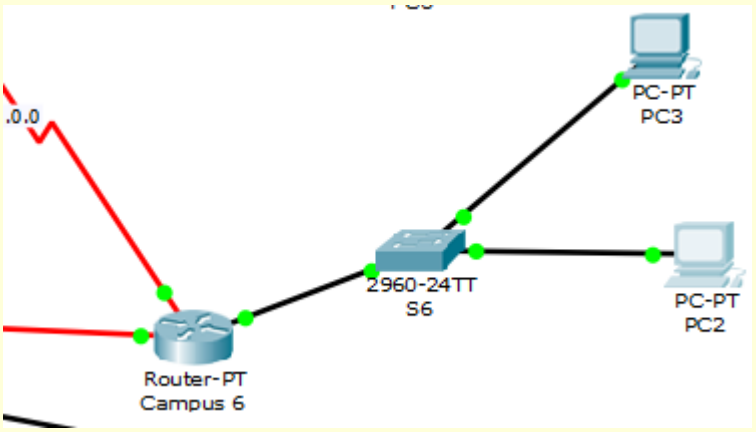
Campus 5

```
interface fa0/0
ip address 66.0.0.254 255.0.0.0
no shut
do wr
exit
interface se2/0
ip address 150.0.0.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 151.0.0.2 255.255.0.0
no shut
do wr
exit
interface se6/0
ip address 152.0.0.2 255.255.0.0
no shut
do wr
exit
```



Campus 6

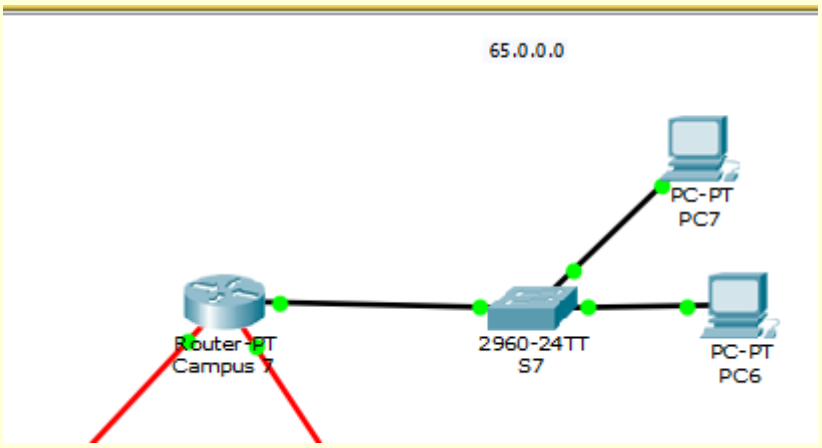
```
interface fa0/0
ip address 64.0.0.254 255.0.0.0
no shut
do wr
exit
interface se2/0
ip address 158.0.0.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 159.0.0.1 255.255.0.0
no shut
do wr
Exit
```



Campus 7

```
interface fa0/0
ip address 65.0.0.254 255.0.0.0
no shut
do wr
exit
interface se2/0
ip address 157.0.0.1 255.255.0.0
clock rate 64000
```

```
no shut
do wr
exit
interface se3/0
ip address 159.0.0.2 255.255.0.0
clock rate 64000
no shut
do wr
exit
```



Campus 8

```
interface fa0/0
ip address 164.0.0.254 255.255.0.0
no shut
do wr
exit
interface se2/0
ip address 156.0.0.2 255.255.0.0
no shut
do wr
exit
interface se3/0
ip address 155.0.0.2 255.255.0.0
no shut
do wr
exit
interface se6/0
ip address 150.0.0.2 255.255.0.0
no shut
do wr
exit
interface se7/0
ip address 158.0.0.2 255.255.0.0
no shut
do wr
exit
interface se8/0
ip address 157.0.0.2 255.255.0.0
no shut
do wr
exit
```

OSPF ALGORITHM

Campus 1

```
router ospf 1
network 154.0.0.0 0.0.255.255 area 1
network 156.0.0.0 0.0.255.255 area 1
network 60.0.0.0 0.255.255.255 area 1
exit
```

Campus 2

```
router ospf 2
network 154.0.0.0 0.0.255.255 area 1
network 155.0.0.0 0.0.255.255 area 1
network 61.0.0.0 0.255.255.255 area 1
exit
```

Campus 3

```
router ospf 3
network 151.0.0.0 0.0.255.255 area 1
network 153.0.0.0 0.0.255.255 area 1
network 62.0.0.0 0.255.255.255 area 1
exit
```

Campus 4

```
router ospf 4
network 152.0.0.0 0.0.255.255 area 1
network 153.0.0.0 0.0.255.255 area 1
network 63.0.0.0 0.255.255.255 area 1
exit
```

Campus 5

```
router ospf 5
network 150.0.0.0 0.0.255.255 area 1
network 151.0.0.0 0.0.255.255 area 1
network 152.0.0.0 0.0.255.255 area 1
network 66.0.0.0 0.255.255.255 area 1
exit
```

Campus 6

```
router ospf 6
network 158.0.0.0 0.0.255.255 area 1
network 159.0.0.0 0.0.255.255 area 1
network 64.0.0.0 0.255.255.255 area 1
exit
```

Campus 7

```
router ospf 7
```

```
network 157.0.0.0 0.0.255.255 area 1
network 159.0.0.0 0.0.255.255 area 1
network 65.0.0.0 0.255.255.255 area 1
exit
```

Campus 8

```
router ospf 8
network 150.0.0.0 0.0.255.255 area 1
network 155.0.0.0 0.0.255.255 area 1
network 156.0.0.0 0.0.255.255 area 1
network 157.0.0.0 0.0.255.255 area 1
network 158.0.0.0 0.0.255.255 area 1
network 164.0.0.0 0.0.255.255 area 1
exit
```

DHCP POOL Capture

Campus 1

```
ip dhcp pool LAN1
network 60.0.0.0 255.0.0.0
default-router 60.0.0.254
dns-server 164.0.0.2
exit
```

Campus 2

```
ip dhcp pool LAN2
network 61.0.0.0 255.0.0.0
default-router 61.0.0.254
dns-server 164.0.0.2
exit
```

Campus 3

```
ip dhcp pool LAN3
network 62.0.0.0 255.0.0.0
default-router 62.0.0.254
dns-server 164.0.0.2
exit
```

Router 4

```
ip dhcp pool LAN4
network 63.0.0.0 255.0.0.0
default-router 63.0.0.254
dns-server 164.0.0.2
exit
```

Campus 5

```
ip dhcp pool LAN5
network 66.0.0.0 255.0.0.0
default-router 66.0.0.254
dns-server 164.0.0.2
exit
```

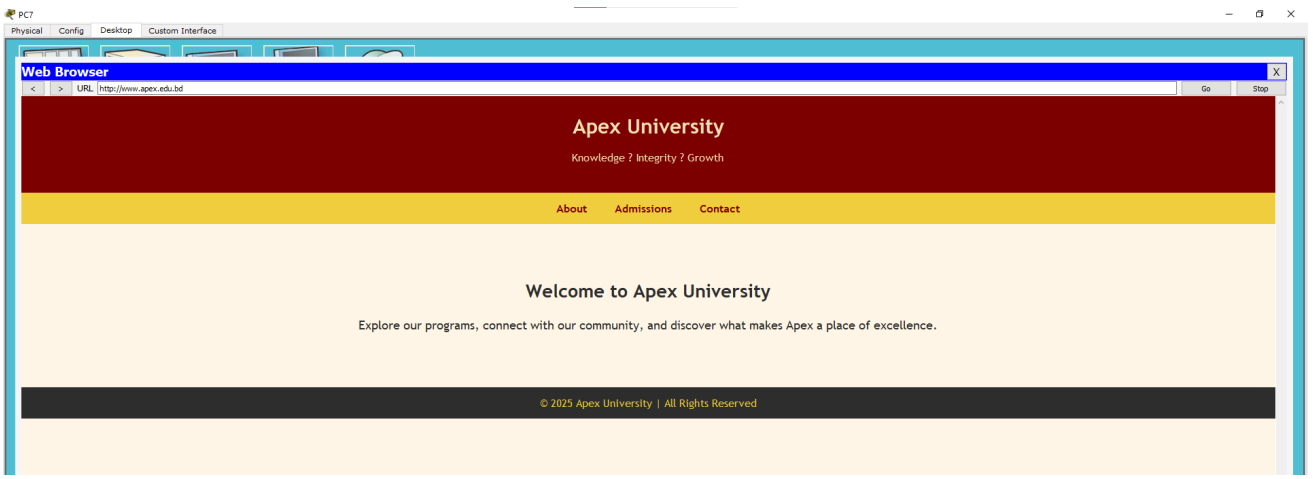
Campus 6

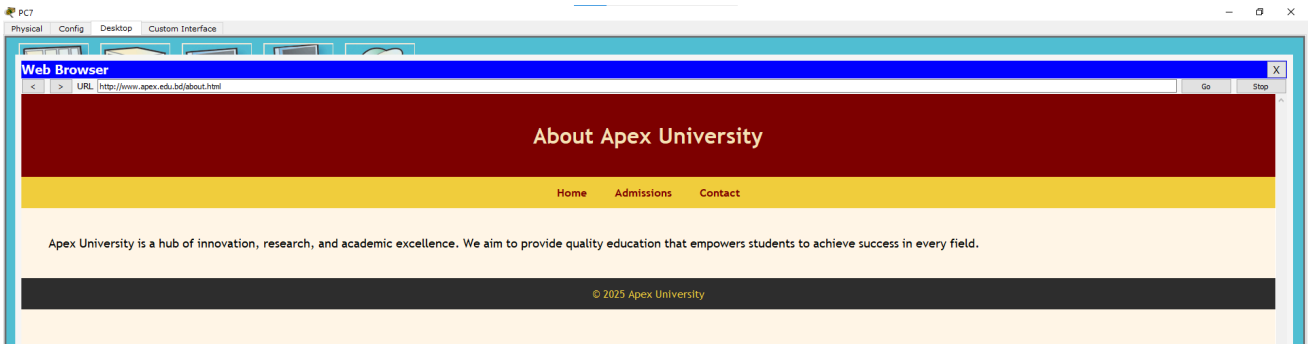
```
ip dhcp pool LAN6
network 64.0.0.0 255.0.0.0
default-router 64.0.0.254
dns-server 164.0.0.2
exit
```

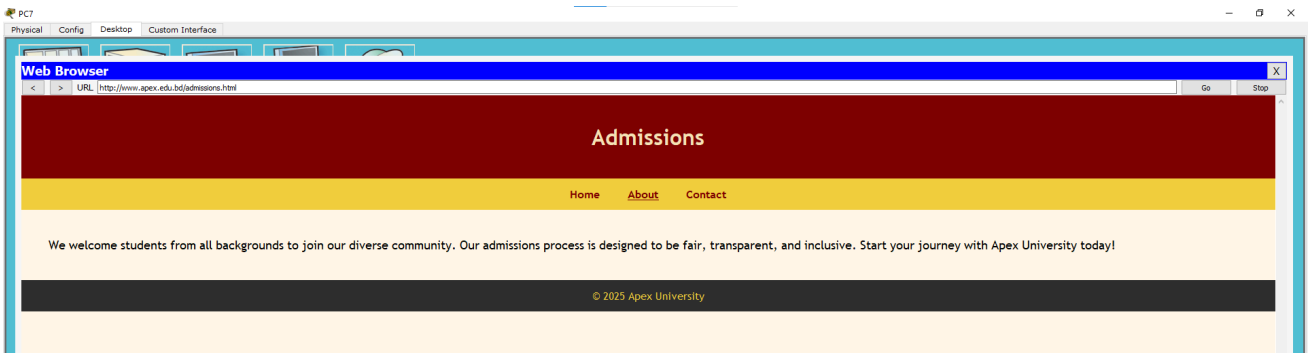
Campus 7

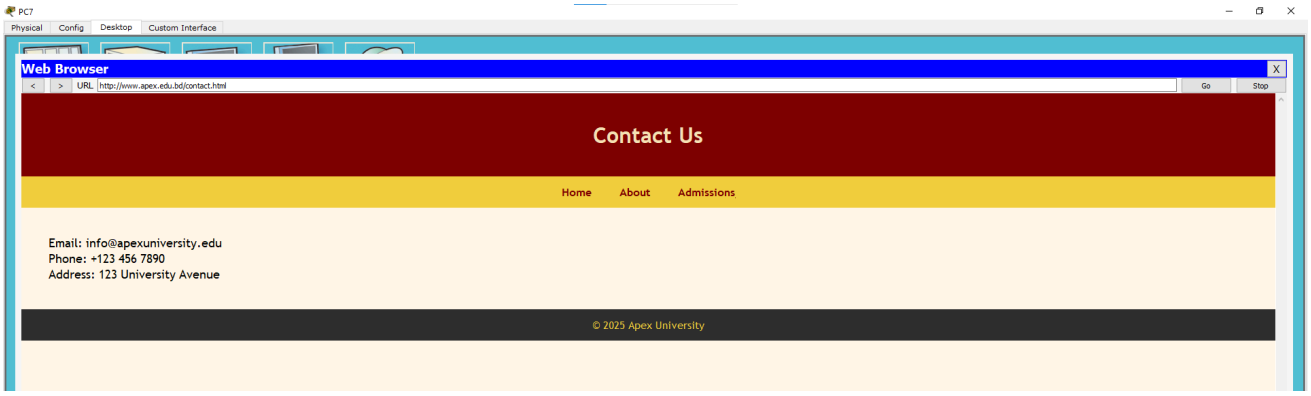
```
ip dhcp pool LAN7
network 65.0.0.0 255.0.0.0
default-router 65.0.0.254
dns-server 164.0.0.2
exit
```

Website :

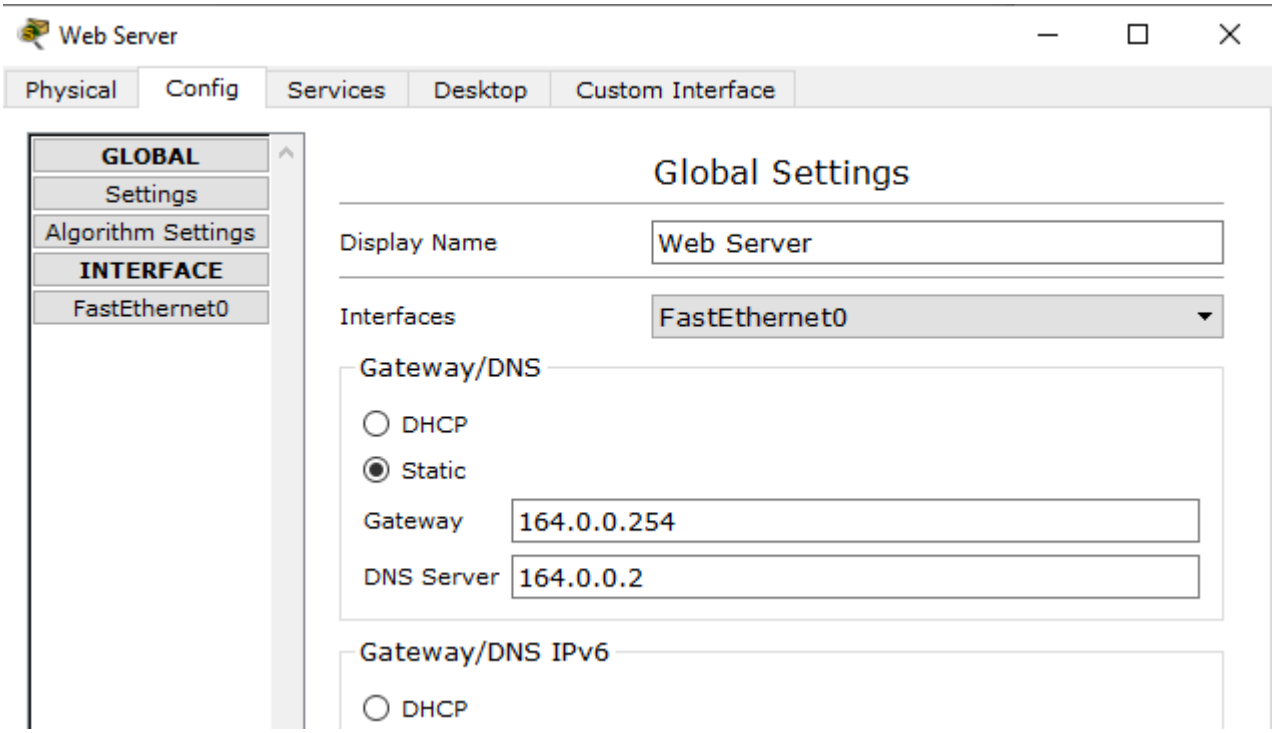

PC7
Physical
Config
Desktop
Custom Interface
Web Browser
URL
http://www.apex.edu.bd
Go
Stop
Apex University
Knowledge ? Integrity ? Growth
About
Admissions
Contact
Welcome to Apex University
Explore our programs, connect with our community, and discover what makes Apex a place of excellence.
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Physical
Config
Desktop
Custom Interface
Web Browser
URL
http://www.apex.edu.bd/about.html
Go
Stop
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Home
Admissions
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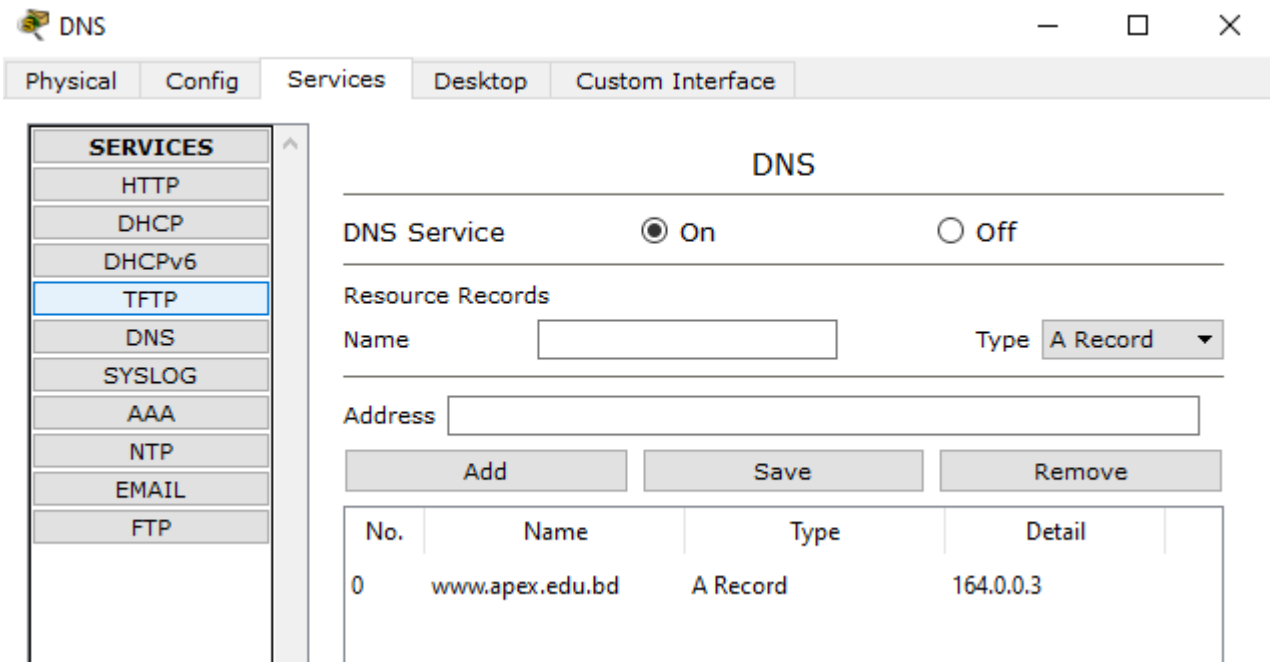

PC7
Physical
Config
Desktop
Custom Interface
Web Browser
URL
http://www.apex.edu.bd/admissions.html
Go
Stop
Admissions
Home
About
Contact
We welcome students from all backgrounds to join our diverse community. Our admissions process is designed to be fair, transparent, and inclusive. Start your journey with Apex University today!
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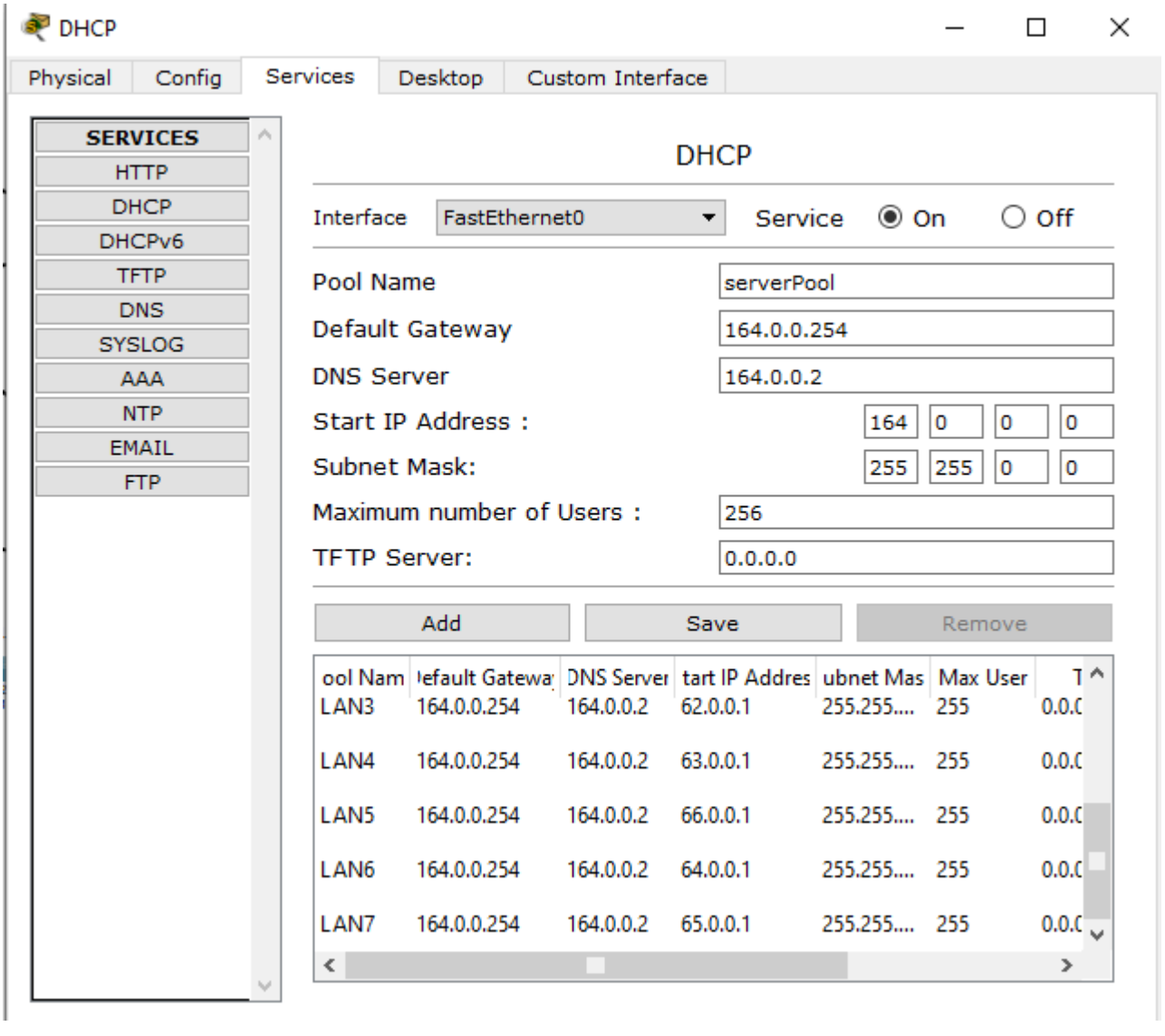
Webserver :



DNS Server :


DNS
Physical
Config
Services
Desktop
Custom Interface
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0
www.apex.edu.bd
A Record
164.0.0.3

DHCP Server :


DHCP
Physical
Config
Services
Desktop
Custom Interface
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
164.0.0.254
DNS Server
164.0.0.2
Start IP Address :
164
0
0
0
Subnet Mask:
255
255
0
0
Maximum number of Users :
256
TFTP Server:
0.0.0.0
Add
Save
Remove
ool Nam
efault Gatewa
DNS Server
tart IP Addres
ubnet Mas
Max User
LAN3
164.0.0.254
164.0.0.2
62.0.0.1
255.255....
255
LAN4
164.0.0.254
164.0.0.2
63.0.0.1
255.255....
255
LAN5
164.0.0.254
164.0.0.2
66.0.0.1
255.255....
255
LAN6
164.0.0.254
164.0.0.2
64.0.0.1
255.255....
255
LAN7
164.0.0.254
164.0.0.2
65.0.0.1
255.255....
255

Conclusion

The proposed network design for Apex University establishes a secure, resilient, and future-ready infrastructure. By leveraging dynamic routing, DHCP, and an efficient subnetting strategy, the architecture ensures seamless communication, optimized resource usage, and simplified administration. This foundation not only supports current academic and administrative operations but also positions the university to adapt effectively to future technological advancements and growth.